1. A simple sentence consists of:

(A) Two independent clauses
(B) One independent clause
(C) One dependent clause
(D) A dependent and independent clause

2. Identify the compound sentence among the following options:

(A) The enzyme dissolved.
(B) The enzyme dissolved, but it denatured.
(C) When the enzyme dissolved, it acted.
(D) Dissolving quickly, the enzyme reacted.

3. A complex sentence is composed of:

(A) Two independent clauses
(B) One independent + one or more dependent
(C) Only dependent clauses
(D) Two coordinating clauses

4. Which sentence represents a compound-complex structure?

(A) The pump failed, so the test stopped.
(B) Because the pump failed, the test stopped.
(C) Because the pump failed, the run stopped, and data was lost.
(D) The pump failed rapidly.

5. What is the subject of: 'In the sterile incubator grew three vibrant colonies'?

(A) incubator
(B) sterile
(C) colonies
(D) grew

6. Identify the independent clause in: 'Although the culture was heated, the spores survived.'

(A) Although the culture was heated
(B) the spores survived
(C) Although the culture
(D) was heated

7. A sentence fragment is defined as:

(A) A clause with a comma splice
(B) A grammatically incomplete sentence
(C) A run-on sentence
(D) A complex sentence

8. Identify the sentence fragment:

(A) The cells were harvested.
(B) Because the temperature dropped suddenly.
(C) The yield exceeded expectations.
(D) Spectrophotometry confirmed purity.

9. A comma splice occurs when two independent clauses are joined by:

(A) A semicolon
(B) A comma without a coordinator
(C) A coordinating conjunction alone
(D) A colon

10. Which punctuation mark can join two independent clauses without a conjunction?

(A) Comma
(B) Semicolon
(C) Hyphen
(D) Slash

11. Identify the fused (run-on) sentence:

(A) The broth boiled it evaporated.
(B) The broth boiled, and it evaporated.
(C) The broth boiled; it evaporated.
(D) When the broth boiled, it evaporated.

12. What is the predicate in: 'The laboratory technician calibrated the micropipettes'?

(A) The laboratory technician
(B) calibrated the micropipettes
(C) micropipettes
(D) laboratory

13. In 'The protein denatured', what type of clause is this?

(A) Dependent clause
(B) Independent clause
(C) Relative clause
(D) Noun clause

14. Choose the correct question tag: 'The PCR reaction succeeded, ___?'

(A) did it
(B) didn't it
(C) wasn't it
(D) hasn't it

15. Choose the correct question tag: 'The samples have not degraded, ___?'

(A) have they
(B) haven't they
(C) did they
(D) weren't they

16. Choose the correct question tag: 'Let us begin the gel electrophoresis, ___?'

(A) will we
(B) shall we
(C) do we
(D) should we

17. Which conjunction joins independent clauses in a compound sentence?

(A) FANBOYS coordinating conjunction
(B) Subordinating conjunction
(C) Relative pronoun
(D) Correlative adverb

18. Identify the noun clause: 'That the plasmid was inserted correctly was proven by sequencing.'

(A) That the plasmid was inserted correctly
(B) was proven by sequencing
(C) by sequencing
(D) the plasmid was inserted

19. Identify the adverbial clause: 'After the centrifugation completed, we decanted the supernatant.'

(A) After the centrifugation completed
(B) we decanted the supernatant
(C) the supernatant
(D) decanted the supernatant

20. What is the grammatical role of an adjective clause?

(A) Modify a verb
(B) Modify a noun or pronoun
(C) Act as subject of the clause
(D) Connect two sentences

21. Choose the correct verb: 'Neither the PI nor the graduate students ___ present.'

(A) was
(B) were
(C) is
(D) has been

22. Choose the correct verb: 'Neither the graduate students nor the PI ___ present.'

(A) was
(B) were
(C) are
(D) have been

23. In subject-verb agreement, intervening phrases like 'together with':

(A) Make the subject plural
(B) Do not affect subject number
(C) Turn verbs into gerunds
(D) Require plural past tense

24. Select the sentence with correct subject-verb concord:

(A) The lead researcher, along with two assistants, is preparing the assay.
(B) The lead researcher, along with two assistants, are preparing the assay.
(C) The lead researcher, along with two assistants, were preparing the assay.
(D) The lead researcher, along with two assistants, have prepared the assay.

25. What is the grammatical term for a sentence with two subjects joined by 'and'?

(A) Complex subject
(B) Compound subject
(C) Split subject
(D) Subordinate subject

26. Identify the sentence with a compound predicate:

(A) The scientist thawed the sample and loaded the gel.
(B) The scientist thawed the sample, and the technician loaded the gel.
(C) Because the scientist thawed the sample, it worked.
(D) The scientist and the technician worked.

27. Which of the following is an imperative sentence used in laboratory protocols?

(A) You should centrifuge the tube.
(B) Centrifuge the tube at 10,000 g.
(C) Can you centrifuge the tube?
(D) The tube was centrifuged.

28. An exclamatory sentence in informal speech typically ends with:

(A) A question mark
(B) An exclamation mark
(C) A colon
(D) A semicolon

29. In technical scientific reports, which sentence mood is overwhelmingly dominant?

(A) Imperative mood
(B) Subjunctive mood
(C) Indicative mood
(D) Optative mood

30. Choose the correct question tag: 'Nobody witnessed the contamination, ___?'

(A) did they
(B) didn't they
(C) did he
(D) was it

31. Choose the correct question tag: 'The autoclave is fully functional, ___?'

(A) is it
(B) isn't it
(C) wasn't it
(D) doesn't it

32. Identify the direct object: 'The restriction endonuclease cleaved the circular plasmid.'

(A) The restriction endonuclease
(B) cleaved
(C) the circular plasmid
(D) circular

33. Identify the indirect object: 'The vendor provided the biotechnology department new reagents.'

(A) The vendor
(B) provided
(C) the biotechnology department
(D) new reagents

34. A sentence exhibiting inverted syntax begins with:

(A) The subject
(B) An adverbial or predicate element
(C) A direct object
(D) A coordinating conjunction

35. In 'Rarely do such mutations survive', what caused the inversion?

(A) The auxiliary verb do
(B) The initial negative adverb Rarely
(C) The plural noun mutations
(D) The transitive verb survive

36. Identify the coordinating conjunction expressing contrast:

(A) And
(B) Or
(C) But
(D) So

37. Identify the coordinating conjunction expressing causal consequence:

(A) For
(B) Nor
(C) Yet
(D) So

38. Choose the correct question tag: 'You have calibrated the pH meter, ___?'

(A) have you
(B) haven't you
(C) did you
(D) didn't you

39. In 'The findings are consistent', the word 'consistent' functions as:

(A) Predicate adjective (subject complement)
(B) Direct object
(C) Indirect object
(D) Objective complement

40. Which sentence structure is best for stating concise experimental conclusions?

(A) Complex-compound with passive stacks
(B) Clear, direct simple or complex sentence
(C) Run-on sentence with multiple FANBOYS
(D) A series of sentence fragments

41. Identify the sentence containing a misplaced modifier:

(A) The technician observed the dividing cells using a high-resolution microscope.
(B) Using a high-resolution microscope, the technician observed the dividing cells.
(C) The technician observed using a microscope the cells dividing.
(D) Cells dividing were observed by the technician.

42. Select the sentence free from dangling modifiers:

(A) Walking into the lab, the alarm sounded loudly.
(B) Walking into the lab, the technician heard the alarm.
(C) Walking into the lab, the spill was detected.
(D) Walking into the lab, an odor was smelled.

43. Identify the sentence that violates parallel structure in an academic listing:

(A) The protocol requires washing, centrifuging, and to dry the pellet.
(B) The protocol requires washing, centrifuging, and drying the pellet.
(C) The protocol requires that we wash, centrifuge, and dry the pellet.
(D) The protocol requires washing, centrifugation, and drying.

44. Choose the correct verb: 'Ten milliliters of buffer solution ___ added to the vessel.'

(A) were
(B) was
(C) are
(D) have been

45. Choose the correct verb: 'A total of twenty mice ___ utilized in the preliminary trial.'

(A) was
(B) were
(C) is
(D) has been

46. Choose the correct verb: 'The number of contaminated cultures ___ negligible.'

(A) were
(B) was
(C) are
(D) have been

47. Choose the correct verb: 'A number of mutant colonies ___ detected on the agar plate.'

(A) was
(B) were
(C) is
(D) has been

48. Identify the sentence with correct concord for proportional expressions:

(A) Two-thirds of the biomass were lost during filtration.
(B) Two-thirds of the biomass was lost during filtration.
(C) Two-thirds of the biomass are lost during filtration.
(D) Two-thirds of the biomass have been lost.

49. Identify the sentence with correct concord for countable proportions:

(A) Fifty percent of the clones was sequenced.
(B) Fifty percent of the clones were sequenced.
(C) Fifty percent of the clones is sequenced.
(D) Fifty percent of the clones has been sequenced.

50. Which sentence demonstrates negative inversion correctly?

(A) Seldom the enzyme exhibited such high activity.
(B) Seldom did the enzyme exhibit such high activity.
(C) Seldom the enzyme did exhibit such high activity.
(D) Seldom has exhibit the enzyme such high activity.

51. Identify the sentence with faulty predication:

(A) The purpose of the assay is when we quantify DNA.
(B) The purpose of the assay is to quantify DNA.
(C) The assay is intended for DNA quantification.
(D) We designed the assay to quantify DNA.

52. Select the correct punctuation to fix the comma splice: 'The primer annealed, the polymerase extended.'

(A) The primer annealed; the polymerase extended.
(B) The primer annealed the polymerase extended.
(C) The primer annealed, so the polymerase extended.
(D) Both A and C are grammatically correct.

53. In 'What we discovered during the screening altered our hypothesis', the underlined clause is a:

(A) Noun clause acting as subject
(B) Adverbial clause of time
(C) Relative adjective clause
(D) Independent coordinate clause

54. Choose the correct question tag: 'Nothing went wrong during the incubation cycle, ___?'

(A) did it
(B) didn't it
(C) was it
(D) weren't they

55. Choose the correct question tag: 'I am authorized to handle biohazardous pathogens, ___?'

(A) am I not
(B) aren't I
(C) ain't I
(D) Both A and B are acceptable

56. Identify the sentence with a squinting modifier:

(A) Students who prepare carefully frequently achieve positive results.
(B) Students who prepare carefully will achieve positive results frequently.
(C) Students who prepare carefully often achieve positive results.
(D) Students frequently who prepare carefully achieve results.

57. Which sentence correctly uses an absolute phrase (nominative absolute)?

(A) The reaction having reached equilibrium, we recorded absorbance.
(B) Reaching equilibrium, absorbance was recorded.
(C) Absorbance was recorded having reached equilibrium.
(D) Having reached equilibrium, absorbance recorded.

58. Identify the sentence with a non-restrictive relative clause correctly punctuated:

(A) Enzymes, that are biocatalysts accelerate reactions.
(B) Enzymes that are biocatalysts, accelerate reactions.
(C) Enzymes, which are biocatalysts, accelerate reactions.
(D) Enzymes which are biocatalysts accelerate reactions.

59. Choose the correct verb: 'Every flask, tube, and beaker ___ sterilized prior to the procedure.'

(A) were
(B) was
(C) are
(D) have been

60. Choose the correct verb: 'Bread and butter ___ provided as standard sustenance.'

(A) is
(B) are
(C) were
(D) have been

61. Choose the correct verb: 'The committee ___ divided on whether to approve human trials.'

(A) is
(B) was
(C) were
(D) has been

62. Select the sentence with parallel correlative conjunctions:

(A) Either the pipette is uncalibrated or the buffer is expired.
(B) Either the pipette is uncalibrated or expired buffer.
(C) The pipette is either uncalibrated or the buffer expired.
(D) Either uncalibrated pipette or buffer expired.

63. Identify the sentence where the subject and predicate complement agree in meaning:

(A) The most critical factor is the lack of proper reagents.
(B) The most critical factor are reagents.
(C) The most critical factors is the lack of reagents.
(D) The factors is lack of reagents.

64. In 'Hardly had the thermal cycler reached 95 deg C when the power failed', 'Hardly...when' expresses:

(A) Strict chronological immediacy with inversion
(B) Causal correlation
(C) Concessive contrast
(D) Conditional contingency

65. Choose the correct question tag: 'There are forty wells remaining on the microtiter plate, ___?'

(A) aren't there
(B) isn't there
(C) aren't they
(D) don't they

66. Identify the dependent clause type in: 'The hypothesis that telomerase prevents senescence is validated.'

(A) Noun complement clause (appositive)
(B) Adverbial clause of condition
(C) Coordinate clause
(D) Independent clause

67. Choose the correct verb: 'The criteria for selecting the candidate vector ___ stringent.'

(A) was
(B) were
(C) is
(D) has been

68. Choose the correct verb: 'The phenomenon of RNA interference ___ observed across eukaryotic taxa.'

(A) is
(B) are
(C) were
(D) have been

69. Which sentence properly avoids a run-on using a subordinating conjunction?

(A) The cell burst the osmotic pressure was too high.
(B) The cell burst because the osmotic pressure was too high.
(C) The cell burst, the osmotic pressure was high.
(D) The cell burst; because pressure was high.

70. Identify the sentence with an appositive phrase correctly punctuated:

(A) CRISPR-Cas9 a revolutionary gene editing tool was utilized.
(B) CRISPR-Cas9, a revolutionary gene editing tool, was utilized.
(C) CRISPR-Cas9 a revolutionary gene editing tool, was utilized.
(D) CRISPR-Cas9, a revolutionary gene editing tool was utilized.

71. Select the sentence with correct syntax in comparative clauses:

(A) The enzyme has greater affinity for substrate A than substrate B.
(B) The enzyme has greater affinity for substrate A than for substrate B.
(C) The enzyme has greater affinity for substrate A than to substrate B.
(D) The enzyme has greater affinity for substrate A compared substrate B.

72. Identify the elliptical clause in: 'While conducting the electrophoresis, the student wore safety goggles.'

(A) While conducting the electrophoresis
(B) the student wore safety goggles
(C) wore safety goggles
(D) While conducting

73. Choose the correct verb: 'Physics ___ fundamental to understanding biomechanical shear forces.'

(A) are
(B) is
(C) were
(D) have been

74. Choose the correct question tag: 'Few researchers disputed the experimental findings, ___?'

(A) did they
(B) didn't they
(C) do they
(D) haven't they

75. Choose the correct question tag: 'A few researchers disputed the experimental findings, ___?'

(A) did they
(B) didn't they
(C) do they
(D) haven't they

76. Identify the sentence with a dangling infinitive modifier:

(A) To optimize PCR yield, magnesium concentration was adjusted.
(B) To optimize PCR yield, we adjusted magnesium concentration.
(C) In order to optimize PCR yield, researchers adjusted magnesium.
(D) For optimizing PCR yield, magnesium was adjusted by us.

77. Select the sentence that avoids faulty parallelism when comparing clauses:

(A) Synthesizing peptides in vitro is more demanding than to express them in vivo.
(B) Synthesizing peptides in vitro is more demanding than expressing them in vivo.
(C) To synthesize peptides in vitro is more demanding than expressing them in vivo.
(D) Synthesizing peptides in vitro is more demanding than in vivo expression.

78. Which sentence features correct grammatical concord with a compound subject joined by 'as well as'?

(A) The primary investigator, as well as her doctoral candidates, were recognized.
(B) The primary investigator, as well as her doctoral candidates, was recognized.
(C) The primary investigator as well as her doctoral candidates are recognized.
(D) The primary investigator, as well as her doctoral candidates, have been recognized.

79. Identify the sentence that correctly utilizes locative inversion for technical emphasis:

(A) On the upper shelf of the biosafety cabinet was placed the stock virus.
(B) On the upper shelf of the biosafety cabinet the stock virus was placed.
(C) The stock virus was placed on the upper shelf of cabinet.
(D) Placed on the upper shelf was the stock virus of cabinet.

80. Select the sentence that eliminates an awkward nominal stack (noun adjunct pileup):

(A) We utilized a real-time reverse-transcription polymerase chain reaction thermal cycler machine.
(B) We used a thermal cycler designed for real-time RT-PCR assays.
(C) We utilized a real time reverse transcription PCR instrument.
(D) Real-time RT-PCR thermal cycler machine was utilized.

81. Choose the correct verb: 'More than one recombinant plasmid ___ evaluated during the screening.'

(A) were
(B) was
(C) are
(D) have been

82. Choose the correct verb: 'Many a promising biofuel candidate ___ failed to achieve commercial viability.'

(A) have
(B) has
(C) were
(D) are

83. Identify the sentence with a subtle comma splice across transitional phrases:

(A) The cell culture density peaked on day 4, in contrast, cell viability declined sharply.
(B) The cell culture density peaked on day 4; in contrast, cell viability declined sharply.
(C) The cell culture density peaked on day 4 in contrast cell viability declined.
(D) The cell culture density peaked on day 4, however viability declined.

84. Choose the correct question tag: 'She had hardly inoculated the flask before the lights went out, ___?'

(A) hadn't she
(B) had she
(C) did she
(D) didn't she

85. Identify the sentence with correct concord where the subject is a gerund phrase with plural objects:

(A) Purifying monoclonal antibodies ___ several chromatography steps.
(B) require
(C) requires
(D) are requiring

86. Select the sentence with a squinting modifier that impairs semantic precision:

(A) Observing mutant phenotypes regularly yields critical insights.
(B) Regularly observing mutant phenotypes yields critical insights.
(C) Observing mutant phenotypes yields critical insights regularly.
(D) Both B and C eliminate the ambiguity found in A.

87. Identify the sentence demonstrating grammatically flawless correlative parallelism:

(A) The therapeutic drug not only reduces tumor volume but also inhibits metastasis.
(B) The therapeutic drug not only reduces tumor volume but also metastasis is inhibited.
(C) Not only the therapeutic drug reduces tumor volume but inhibits metastasis.
(D) The therapeutic drug reduces not only tumor volume but also inhibits metastasis.

88. Choose the correct verb: 'What appears to be insignificant contaminants ___ actually secondary metabolites.'

(A) is
(B) are
(C) was
(D) has been

89. Select the sentence that avoids an unattached dangling participle in scientific methods:

(A) Having stained the nitrocellulose membrane, the band pattern was analyzed.
(B) Having stained the nitrocellulose membrane, we analyzed the band pattern.
(C) After staining the nitrocellulose membrane, the band pattern appeared.
(D) The membrane having been stained, band pattern analyzed.

90. In 'No sooner had the enzyme been introduced ___ the substrate initiated turnover':

(A) than
(B) when
(C) then
(D) before

91. Identify the sentence that correctly resolves an ambiguous antecedent pronoun:

(A) The technician placed the conical flask on the shaker, and it shattered.
(B) The conical flask shattered when the technician placed it on the shaker.
(C) When the technician placed the flask on the shaker, it broke.
(D) Placing the flask on the shaker, it shattered.

92. Choose the correct verb: 'The majority of the committee ___ in favor of approving the recombinant DNA protocol.'

(A) is
(B) are
(C) were
(D) has been

93. Choose the correct verb: 'A majority of the surveyed laboratory personnel ___ personal protective equipment.'

(A) wear
(B) wears
(C) is wearing
(D) has worn

94. Identify the sentence with an error in predicate agreement after 'neither of':

(A) Neither of the two candidate vectors was capable of replication.
(B) Neither of the two candidate vectors were capable of replication.
(C) Neither candidate vector was capable of replication.
(D) Neither vector replicated in the host cell.

95. Select the sentence that correctly preserves grammatical parallelism across compound infinitives:

(A) We sought to isolate the protein, sequence its amino acids, and determining its structure.
(B) We sought to isolate the protein, to sequence its amino acids, and to determine its structure.
(C) We sought to isolate the protein, sequence its amino acids, and determine its structure.
(D) Both B and C are grammatically parallel and acceptable.

96. Choose the correct question tag: 'Everyone understands the biosafety level 3 decontamination procedures, ___?'

(A) doesn't he
(B) don't they
(C) doesn't everyone
(D) aren't they

97. Identify the sentence with an ungrammatical double subordinate conjunction:

(A) Because the culture was contaminated, so we discarded it.
(B) Because the culture was contaminated, we discarded it.
(C) Since the culture was contaminated, so we discarded it.
(D) Although the culture was cloudy, but we tested it.

98. Choose the correct verb: 'Species richness, together with functional diversity, ___ ecosystem resilience.'

(A) dictate
(B) dictates
(C) are dictating
(D) have dictated

99. Select the sentence with flawless subject-verb agreement involving a cleft sentence:

- **(A)** It is the altered glycosylation patterns that ___ responsible for antibody clearance.
- **(B)** is
- **(C)** are
- **(D)** was

100. Identify the syntactically flawless sentence adhering to formal scientific syntax:

- **(A)** Not only did the CRISPR guide RNA target the oncogene with high fidelity, but it also produced minimal off-target cleavage.
- **(B)** Not only the CRISPR guide RNA targeted the oncogene with high fidelity, but also produced minimal off-target cleavage.
- **(C)** The CRISPR guide RNA not only did target the oncogene, but also minimal off-target cleavage was produced.
- **(D)** Not only did the CRISPR guide RNA target the oncogene with high fidelity, but minimal off-target cleavage.

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Classify the four fundamental sentence architectural types in English syntax and discuss their strategic deployment in biotechnology manuscripts.

Q2. Analyze the syntax and mechanics of dependent clauses (noun, adjective, and adverbial clauses) in biomedical research protocols.

Q3. Examine the grammatical pathology of sentence fragments and demonstrate three systematic remediation techniques in technical reports.

Q4. Analyze comma splices and fused sentences in academic prose, detailing four grammatically sound correction methods.

Q5. Examine the syntax of misplaced, dangling, and squinting modifiers in laboratory protocols, providing unambiguous revisions.

Q6. Detail the principle of grammatical and syntactic parallelism in research article methodologies, results, and headings.

Q7. Analyze subject-verb agreement rules when subjects are modified by parenthetical prepositional phrases or quasi-conjunctions.

Q8. Examine subject-verb concord with correlative conjunctions (either...or, neither...nor, not only...but also) and the Rule of Proximity.

Q9. Analyze concord rules for indefinite quantifiers, fractions, percentages, and mass nouns in biochemical data reporting.

Q10. Discuss inverted sentence structures (negative inversion and locative inversion) and their stylistic impact in technical discourse.

Q11. Examine the formation, polarity, and grammatical rules of question tags in professional communication and defense vivas.

Q12. Analyze the syntax of absolute phrases (nominative absolutes) and their economy in condensing scientific descriptions.

Q13. Examine the grammatical structure of noun clauses, appositive clauses, and their punctuation in scientific claims.

Q14. Analyze adverbial clauses of condition, concession, and cause, focusing on punctuation conventions and register.

Q15. Examine the grammatical rules of cleft sentences ('it'-clefts and 'wh'-clefts) for establishing empirical scientific emphasis.

Q16. Detail the syntax of correlative conjunctions and analyze errors in parallel structures involving scientific comparisons.

Q17. Analyze the syntax and punctuation of appositive phrases in defining technical terms and novel biotechnology reagents.

Q18. Examine the grammatical problem of faulty predication in technical writing, illustrating errors and clear corrections.

Q19. Analyze inverted conditional sentences (omission of 'if' with had, were, should) in formal scientific hypotheses.

Q20. Examine noun adjunct pileups (nominal stacks) in biotechnology literature and demonstrate syntactic unpacking techniques.

Q21. Analyze the syntax and punctuation of coordinating conjunctions connecting compound sentences versus compound predicates.

Q22. Discuss subject-verb concord in inverted locative and existential 'there' constructions in scientific descriptions.

Q23. Examine the syntax of elliptical clauses and the hazard of dangling elliptical clauses in technical methodology.

Q24. Detail the rules governing predicate complements, distinguishing predicate nouns, predicate adjectives, and objective complements.

Q25. Analyze parenthetical elements, dash insertions, and brackets in complex scientific syntax.

Q26. Examine the grammatical structure of negative sentences and the avoidance of double negatives in scientific discourse.

Q27. Analyze the syntax of restrictive versus non-restrictive participial phrases in experimental results.

Q28. Examine the rules governing verb concord with subjects modified by 'each', 'every', 'many a', and 'more than one'.

Q29. Analyze the syntactic criteria for forming grammatically valid compound-complex sentences in academic literature.

Q30. Examine common syntactic errors in question-tag formations involving indefinite pronouns and semi-negative adverbs.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [B]
Q2: [B]
Q3: [B]
Q4: [C]
Q5: [C]
Q6: [B]
Q7: [B]
Q8: [B]
Q9: [B]
Q10: [B]
Q11: [A]
Q12: [B]
Q13: [B]
Q14: [B]
Q15: [A]
Q16: [B]
Q17: [A]
Q18: [A]
Q19: [A]
Q20: [B]
Q21: [B]
Q22: [A]
Q23: [B]
Q24: [A]
Q25: [B]
Q26: [A]
Q27: [B]
Q28: [B]
Q29: [C]
Q30: [A]
Q31: [B]
Q32: [C]
Q33: [C]
Q34: [B]
Q35: [B]
Q36: [C]
Q37: [D]
Q38: [B]
Q39: [A]
Q40: [B]
Q41: [A]
Q42: [B]
Q43: [A]
Q44: [B]
Q45: [B]
Q46: [B]
Q47: [B]
Q48: [B]
Q49: [B]
Q50: [B]
Q51: [A]
Q52: [D]
Q53: [A]
Q54: [A]
Q55: [D]
Q56: [A]
Q57: [A]
Q58: [C]
Q59: [B]
Q60: [A]
Q61: [C]
Q62: [A]
Q63: [A]
Q64: [A]
Q65: [A]
Q66: [A]
Q67: [B]
Q68: [A]
Q69: [B]
Q70: [B]
Q71: [B]
Q72: [A]
Q73: [B]
Q74: [A]
Q75: [B]
Q76: [A]
Q77: [B]
Q78: [B]
Q79: [A]
Q80: [B]
Q81: [B]
Q82: [B]
Q83: [B]
Q84: [B]
Q85: [have required]
Q86: [D]
Q87: [A]
Q88: [B]
Q89: [B]
Q90: [A]
Q91: [B]
Q92: [A]
Q93: [A]
Q94: [B]
Q95: [D]
Q96: [B]
Q97: [B]
Q98: [B]
Q99: [has been]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[illegible]

- Adverbial Clauses: Modify verbs, adjectives, or entire clauses, expressing relations of Time (after, before), Cause (because, since), Concession (although, whereas), Condition (if, unless), or Purpose (so that, in order that):
- Example: "The cell lysate was kept on ice so that endogenous proteases would not degrade the target protein."

[illegible]

- "Whereas the wild-type cells grew rapidly, the mutant strain failed to divide."

[illegible]

[Problem Statement]: Analyze subject-verb agreement rules when subjects are modified by parenthetical prepositional phrases or quasi-conjunct
[Detailed Step-by-Step Answer]:
1. Core Grammatical Principle:
The grammatical number of a subject noun is never altered by intervening prepositional phrases, parenthetical expressions, or quasi-connectors:
- Quasi-connectors include: as well as, along with, together with, in addition to, accompanied by, including, except, besides.

2. Diagnostic Case Studies in Biotechnology:
- Example 1: "The lead investigator, along with three postdoctoral researchers, [is / are] attending the genomics symposium."
- Subject: 'The lead investigator' (singular). The parenthetical phrase 'along with...' is non-restrictive.
- Correct: "The lead investigator, along with three postdoctoral researchers, is attending..."
- Example 2: "The gene expression pattern, in addition to protein abundance levels, [correlates / correlate] with clinical outcome."
- Subject: 'The gene expression pattern' (singular).
- Correct: "...correlates with clinical outcome."

3. Plural Compound Subject Exception:
Only the true coordinating conjunction 'and' creates a compound plural subject requiring a plural verb:
- "The lead investigator and three postdoctoral researchers are attending..."

>>Solution to Question 8:

[Problem Statement]: Examine subject-verb concord with correlative conjunctions (either...or, neither...nor, not only...but also) and the Rule of Proximity.

[illegible]

2. Core Syntactic Rules:
- Polarity Reversal:
- Positive Declarative Stem -> Negative Tag:
- "The autoclave reached operating temperature, didn't it?"
- Negative Declarative Stem -> Positive Tag:
- "The reagent has not expired, has it?"
- Auxiliary Matching: The tag must reproduce the exact auxiliary or modal verb present in the stem:
- "The biosensor can detect femtomolar concentrations, can't it?"
- 'Do'-Support: When the stem contains an ordinary lexical verb without an auxiliary, use do/does/did:
- "The bacteria synthesize flagella under starvation, don't they?"
- Pronoun Subject: The tag subject must always be a personal pronoun matching the stem subject.

3. Special Idiomatic Cases:
- Stem with 'I am': Tag is 'aren't I?' (or 'am I not?' in hyper-formal registers): "I am next to defend the thesis, aren't I?"
- Imperative Proposals with 'Let's': Tag is 'shall we?': "Let's review the gel electrophoresis data, shall we?"
- Negative Indefinite Subjects (nobody, nothing, neither): Stem is negative, so tag must be positive: "Nothing contaminated the sterile hood, did it?"

>> Solution to Question 5.12:

Problem Statement: Analyze the syntax of absolute phrases (nominative absolutes) and their function in scientific descriptions.

Object, Predicate Nominative, or Prepositional Complement:
- As Subject: "How the pathogen evades immune recognition remains an open empirical question."
- As Direct Object: "The clinical study proved that the recombinant vaccine elicited neutralizing antibodies."
- As Prepositional Complement: "The results depend on whether the transcription factor binds the promoter."

2. Noun Clauses vs. Restrictive Relative Clauses:
- Noun Complement Clause: Introduces the content of an abstract noun (e.g., fact, hypothesis, finding, idea). 'That' is a conjunction, not a pronoun:
- "The hypothesis that telomerase prevents cellular senescence was supported." (No comma).
- Relative Clause: Modifies an antecedent noun where 'that' functions as a relative pronoun replacing the noun:
- "The hypothesis that was formulated by Dr. Watson was supported."

3. Punctuation Rule:
Restrictive noun clauses and complement clauses are never separated from their governing verbs or nouns by commas.

[illegible]

2. Syntactic Variants:
- 'It'-Cleft: Structure: It + [form of be] + [focused constituent] + [that/who clause]:
- Canonical: "Enzymatic cleavage at site A prevented viral entry."
- Cleft: "It was enzymatic cleavage at site A that prevented viral entry."
- Function: Directly refutes competing hypotheses (e.g., proving that cleavage at site A, not site B, was the causative factor).
- 'Wh'-Cleft (Pseudo-Cleft): Structure: [Wh-clause] + [form of be] + [focused element]:
- "What surprised the investigators was the extraordinary stability of the hyperthermophilic enzyme."

3. Concord in 'It'-Clefts:
The relative clause verb in an 'it'-cleft agrees with the focused constituent, not with the dummy pronoun 'it':
- Correct: "It is the altered post-translational modifications that are responsible for poor receptor binding."

>>Solution to Question 14: [illegible]

- Essential to specifying which entity among several possibilities is meant; no commas:
- "The antibiotic ampicillin was added to select for plasmid retention."
- (Removing 'ampicillin' leaves 'The antibiotic' ambiguous; commas would falsely denote that ampicillin is the only antibiotic).

3. Dash or Parenthesis Appositives for Complex Lists:

When the appositive itself contains internal commas, enclose it with em-dashes (or hyphens in ASCII) or parentheses:

- "Three major cell components - ribosomes, mRNA, and transfer RNA - participate in polypeptide translation."

[illegible]

- Inverted: "Should contamination occur, autoclave the entire batch immediately."

2. Stylistic Impact:
Demonstrates linguistic sophistication, enhances sentence rhythm, and avoids repetitive overuse of the subordinating conjunction 'if' across technical discussions.

[illegible]

[illegible]

[illegible] complements.

nothing, hardly, scarcely, barely).

2. The Double Negative Fallacy:
Pairing two negative words within the same clause cancels the negation and creates an unintended affirmative, which is strictly unacceptable in formal scientific syntax:
- Illogical: "The culture didn't show no signs of contamination."
- Correct: "The culture showed no signs of contamination" OR "The culture did not show any signs of contamination."
- Inadvertent Double Negatives with Semi-Negatives:
- Illogical: "We couldn't hardly detect the fluorescent signal."
- Correct: "We could hardly detect the fluorescent signal."

3. Litotes (Deliberate Understatement in Academic Register):
Using a negative word paired with a negative prefix to express a qualified, cautious positive claim:
- "The difference in binding affinity was not insignificant." (Means the difference was meaningful, but stated with scholarly caution).

>>Solution to Question 23: [illegible] 'more than one'.

- Many a / Many an: An archaic or formal distributive construction that is grammatically singular:
- "Many a candidate compound has failed in preclinical pharmacokinetic trials."
- More than one: Although semantically plural (meaning two or more), its grammatical head is 'one', mandating a singular verb:
- "More than one bacterial colony was discovered on the selective agar plate."
- Contrast: "More than two bacterial colonies were discovered..."

[illegible]

- Correct: "The thermal sensor barely detected the exotherm, did it?"
